

The Second Loop: Durable Correction of Memorized Errors in Frozen Language Models via External Memory, Contrastive Decoding, and Grounded Gatekeeping

Serghei Brinza
Independent Researcher
brinzaengineeringai@gmail.com

August 1, 2026

Abstract

A frozen language model that has memorized a wrong fact repeats it even when the correct fact is placed in its context. We study whether such an error can be corrected *durably* (so that the correction survives a full process reload of the model) without touching a single weight. We introduce the **scar-survival rate**: the fraction of “trap” facts on which an external correction mechanism keeps beating the model’s memorized instinct after N full reloads. On a frozen Qwen2.5-3B-Instruct subject with an independent Qwen2.5-7B-Instruct judge, a per-fact authoritative notebook combined with contrastive decoding turns 0/14 correct answers into 14/14 and holds them at 100% across ten reloads and under stochastic-decoding stress, and at 92.9% under up to +500 entries of notebook ballast. The mechanism has one sharp failure mode: it is *blind to truth*. A single plausible lie placed next to the truth in the notebook drives survival down to 50%. We show that a gatekeeper built from a same-family model cannot close this gap (it inherits the subject’s blind spots, ceiling 64.3%), whereas a gatekeeper grounded in a real external source (live Wikipedia) plus targeted retrieval engineering restores survival to 100%. We validate that the metric ports across three models and two fact domains (100% in all six cells, 540 answers) and that it discriminates a real correction from cosmetic and placebo variants (100% / 0% / 14.3%). Finally, we probe the opposite bound (how rarely external truth can arrive before a self-calibrating system drifts) using a frozen forecaster with per-topic Beta calibration over 1,646 resolved ForecastBench questions swept across eleven feedback schedules; the collapse is binary, with any finite contact (down to once a year) holding and only zero contact failing. We frame the whole arc through Tarski’s undefinability of truth: a model cannot certify its own truthfulness from within, and neither can its copy; a criterion of truth must be supplied from outside. Throughout, we are explicit about the extent to which the headline 100% figures are a consequence of a deterministic setup.

1 Introduction

Instruction-tuned language models carry *memorized instincts*: firmly encoded answers that the model emits with high confidence. When such an instinct is wrong — the tallest mountain measured

Code and data for all six experiments: scar-survival (Part 1), external-grounding (Part 2), thin-channel (Part 3).
Interactive result viewers: huggingface.co/spaces/Laborator/second-loop.

base-to-peak, the first animals sent to space, the exact wording of a famous film line — the model repeats the error even when the correct fact is available to it in context. The standard remedy, weight editing or fine-tuning, changes the object we are trying to study and raises its own durability questions.

This paper asks a narrower and, we argue, more fundamental question. Suppose the model is a *frozen subject*: its weights never change. Can we correct a memorized error with an external mechanism, and will that correction be *durable*: specifically, will it survive a full process reload of the model, or is it only a cosmetic effect that vanishes with process memory? We call a durable correction a **scar**, and we call the whole research program the **Second Loop**: correction that closes back on itself and holds across reloads.

We make the following contributions.

1. **A metric.** The scar-survival rate (Section 3): the fraction of working traps on which the mechanism keeps beating the instinct across N full reloads, scored by an independent semantic judge over the actually generated answer.
2. **A mechanism and its failure mode** (Section 4). A per-fact notebook plus contrastive decoding lifts a frozen 3B model from 0/14 to 14/14 and holds 100% across ten reloads, but collapses to 50% when the notebook is polluted with a plausible lie. The mechanism is blind to truth.
3. **Where the truth criterion must live** (Section 5). A gatekeeper made from a same-family model caps at 64.3%: it shares the subject’s blind spots. Replacing it with a real external source (live Wikipedia) plus targeted retrieval engineering restores 100%. We connect this to Tarski’s undefinability of truth.
4. **Metric validation** (Section 6). The metric ports across three models and two fact domains (100% in all six cells) and discriminates a real scar from cosmetic and placebo controls (100% / 0% / 14.3%).
5. **The opposite bound** (Section 7). Using resolved forecasting questions as a stream of “truth from the future,” we sweep how rarely external feedback can arrive before a self-calibrating frozen system drifts. The collapse is binary: any finite contact holds; only zero contact fails.

A note on honesty of claims, stated up front because it governs how every number below should be read. Several headline figures are 100%. Under a frozen model, a static notebook on disk, and greedy decoding, a full process reload deterministically reproduces the same state, so 100% survival across reloads is partly a property of that determinism. The contribution lies elsewhere: (i) that the protocol is portable across models and domains (Section 6), (ii) that it discriminates real from fake interventions (Section 6), and (iii) that it exposes and then closes a real failure mode — memory pollution — with an external source of truth (Section 5).

2 Related Work

Knowledge editing. Methods such as ROME [12] and MEMIT [13] edit facts by modifying weights. Our setting is deliberately complementary: the weights are frozen and the correction lives entirely outside the model, which lets us ask the durability question without confounding it with weight-update stability. Beyond single-edit stability, sequential editing degrades the edited model [5] and edits ripple unpredictably into related facts [4], an added motivation for keeping corrections outside the weights entirely.

Knowledge conflicts and context faithfulness. When an in-context fact contradicts a model’s parametric memory, models frequently revert to the memorized answer. This context–memory conflict is documented from entity-substitution studies [11] through controlled behavioral analyses [20], and is systematized in the survey of Xu et al. [21]. Our traps are a deliberately adversarial instance: fourteen facts on which the subject’s parametric prior confidently beats a correct in-context fact. The failure mode of §4 (the corrector amplifying a plausible distractor) is the inter-context variant of the same conflict, arising when the notebook itself carries contradictory entries.

Contrastive and context-faithful decoding. Contrastive decoding was introduced as a generation-quality objective [10] and later directed at factuality, either by contrasting layers [3] or, closest to us, by contrasting a run with and without the context, as in context-aware decoding [16]; CK-PLUG [2] and DeCK [1] refine this contrast for knowledge editing. Our mechanism is a direct implementation of this family; our contribution is the durability protocol around the decoder: reload survival, stress loads, and discrimination controls. Our Confidence Gain indicator (Section 4) follows the knowledge-consistency signal of CK-PLUG [2].

Memory corruption in retrieval-augmented systems. Our counterfeit-notebook stress test is a manual instance of corpus poisoning: a handful of injected passages reliably steers retrieval-augmented answers [22, 23], and certified defenses respond by aggregating over isolated passages [19]. Two differences in aim: our injected lie is not optimized (a single plausible decoy suffices to halve survival), and our defense is scored end-to-end by scar survival across reloads rather than by retrieval metrics. The identity of §5 — the defense ceiling equals the verifier’s accuracy — gives this setting a simple predictive law. Broader treatments of these risks place corpus poisoning within a wider trustworthiness agenda for retrieval-augmented systems [14]. Closest in spirit to our gatekeeper, Si et al. [17] pair semantic filtering of the corpus with a conflict-aware stage that screens query, candidate answer and retrieved evidence for contradiction before generation, a design that independently arrives at the same decomposition as our conflict detector plus arbiter, though scored by generation robustness rather than by survival across reloads.

Limits of self-correction. Intrinsic self-correction — a model revising its own output without new information — largely fails on reasoning tasks [6], and surveys locate reliable correction in external feedback [15, 7]. Our clone-arbiter result is the fact-level, mechanistic version of that conclusion: a same-family verifier inherits the subject’s blind spots and caps exactly at its own accuracy (§5), while an external source restores full survival.

Retrieval-augmented grounding. Part 2 grounds a gatekeeper in live Wikipedia retrieval [9]. Our finding there is specifically about the *precision/recall operating point* of the retriever as the binding constraint, not retrieval in general.

Calibration and forecasting. Part 3 uses ForecastBench [8] as a stream of resolved binary questions and applies per-topic Beta calibration.

3 The Scar-Survival Rate

Let T be a set of *working traps*: questions on which the frozen model M , run in its clean regime (no correction mechanism), confidently produces the wrong answer. Let Mem be an external memory of correct facts, each tagged as “truth that overrides the memorized answer”; let Mech be a correction

mechanism running on top of M (contrastive decoding, retrieval augmentation, ...); let N be the number of full process reloads; and let J be an independent semantic judge that labels each answer **correct** or **incorrect** against the reference fact. For each trap t and reload n , $\text{answer}(t, n) = M(\text{Mech}, \text{Mem})(t)$ and $\text{verdict}(t, n) = J(\text{answer}(t, n), \text{gold}(t))$. Then

$$\text{scar-survival rate} = \frac{|\{t \in T : \forall n \in \{1, \dots, N\}, \text{verdict}(t, n) = \text{correct}\}|}{|T|}. \quad (1)$$

Two design choices matter. First, each reload n is a *separate operating system process*: the model is fully unloaded and reloaded from scratch and the notebook is re-read from disk, so a “cosmetic” correction that lives only in process memory cannot survive. Second, scoring is done by a semantic judge over the actual generated answer, not by keyword matching or by a token-level probability metric; Section 4 shows why the latter is unreliable. Throughout this paper $N = 10$ and J is Qwen2.5-7B-Instruct.

4 Experiment 1: The Scar Mechanism

Setup. The frozen subject is Qwen2.5-3B-Instruct, run on a single RTX 3090. We screened both Qwen2.5-7B-Instruct and Qwen2.5-3B-Instruct over the same 58 candidate traps (myths, geography, quotations, niche facts, “false friends”) with the same automatic marker-based classifier, which labels each answer **correct**, **wrong_instinct**, or **ambiguous**. Under that single criterion the two models fire the memorized error at the same rate — $6/58 = 10.3\%$ each — but they differ in how cleanly they get the rest right: the 7B leaves no ambiguous answers, whereas the 3B leaves 7, so $13/58$ (22.4%) of its answers are not unambiguously correct. We chose the 3B because it also hallucinates its own wrong answer readily, which is what the mechanism must override. A manual review of all 58 of the 3B’s answers (not applied to the 7B) yields the **14 working traps** used throughout this paper. Thirteen come from the fired and ambiguous buckets; one (#46) was promoted from an answer the classifier scored **correct** because the generated text contradicts itself arithmetically: it answers “No,” then reports a Venusian day of 243 Earth days against a Venusian year of approximately 225, yet calls the day *shorter* than the year. The external memory is a notebook holding, for each trap, the correct fact.

Correction mechanism. At each decoding step the model is evaluated twice — with the fact in context and without it (pure instinct) — and the next-token distribution is shifted toward the fact:

$$\text{logits} = (1 + \alpha) \cdot \text{logits}_{\text{with fact}} - \alpha \cdot \text{logits}_{\text{instinct}}, \quad \alpha = 1.0, \quad (2)$$

with a plausibility threshold $\beta = 0.1$ that prevents amplifying low-probability nonsense. We also report **Confidence Gain** (after CK-PLUG [2]), the increase in probability mass on the correct answer, $P_{\text{contrastive}} - P_{\text{instinct}}$, as an indicator of override strength.

Three modes. Scored by the independent 7B judge over 14 traps: pure instinct gives 0/14 (0%); the plain notebook (fact in context) gives 14/14 (100%); contrastive decoding also gives 14/14 (100%), with a mean Confidence Gain of +0.744. The distinction between the two 100% columns is qualitative: on trap #37 (“first animal in space”) the fact-in-context barely moved the choice ($P = 0.017$), whereas contrastive raised it to 0.899: the difference between reading off a crib and a genuine intervention.



Figure 1: Scar survival across ten full process reloads (Experiment 1): 14/14 traps held in every cycle.

Table 1: Experiment 1 stress tests (survival by 7B judge over the generated answer).

Load	Survival	Verdict
Baseline (clean, 10 reloads)	100%	—
1. Growing notebook (+50/100/200/500)	92.9%	holds; Gain stable 0.65–0.72
2. Noisy memory (2–3 distractors)	50% (7/14)	fails
3. Stochasticity (temp=0.7, 700 samples)	100%	holds, no spread
4. Sticky instincts ($P_{\text{instinct}} \approx 0$, 12 traps)	100%	holds, Gain +0.58

Scar survival across ten reloads. Running the strict protocol (each cycle a separate process), the scar-survival rate is 100% in every one of the ten cycles (14/14 each time), and the Confidence Gain is bit-for-bit 0.7439 in every cycle (Figure 1). As stated in the introduction, this exactness is a consequence of the deterministic setup; the value of the run is that it establishes the protocol and the baseline against which the stress tests and the discrimination controls (Section 6) are read.

Stress tests. We then probe the override under pressure on the notebook itself (Table 1, Figure 2). It survives a growing notebook (up to +500 ballast entries, 92.9%) and stochastic decoding (temperature 0.7, 700 samples, 100%), and it survives even the stickiest instincts (12 traps with $P_{\text{instinct}} \approx 0$, 100%). It fails under a *noisy notebook*: with two to three plausible distractor entries next to the truth, survival drops to 50% (7/14). Under this load the model copied the false distractor verbatim (“3 seconds,” “K2,” “Play it again, Sam,” “Laika,” “Steamboat Willie”), and on two traps the Confidence Gain went negative.

A methodological caveat. Confidence Gain overestimates robustness under noise. It compares the correct answer against a single canonical wrong one, but free generation can latch onto a *third* distractor outside that comparison (on trap #20 the metric read +0.90 while the model generated “K2”). The honest measure is therefore the semantic judge over the actual answer; we keep Confidence Gain only as an indicator of override strength, not as proof of correctness. This is why the metric in Section 3 is defined over J , not over probabilities.

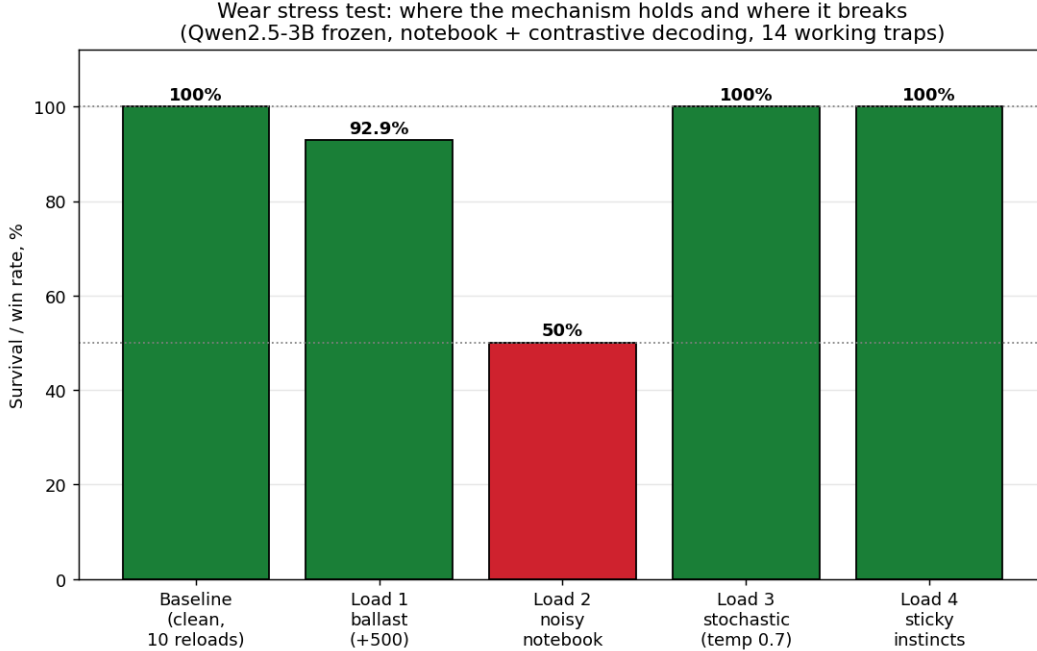


Figure 2: Where the mechanism holds and where it breaks (Experiment 1). Memory pollution, not wear, is the failure mode.

Takeaway. The ceiling of the mechanism is *notebook quality*, not its size, decoding randomness, or instinct strength. Contrastive decoding breaks through the most stubborn instinct, but it will amplify a plausible lie sitting next to the truth just as readily, especially when the lie coincides with the model’s own instinct. This motivates Part 2.

5 Guarding the Notebook: Where the Truth Criterion Must Live

The failure mode of Section 4 is memory pollution. We place a *gatekeeper* at the entrance to the notebook whose job is to reject false or conflicting entries before they reach the corrector. The gatekeeper has three parts: a conflict detector (pairwise comparison of entries for the same question), an arbiter (which adjudicates flagged conflicts), and a quarantine (rejected entries are isolated from the working notebook). The conflict detector caught the contradiction in every noisy entry (14/14); the interesting variable is the *arbiter*.

Guardian 1.0 — a same-family arbiter (Experiment 2). When the arbiter is a second model of the same family (Qwen2.5-7B judging entries for a Qwen2.5-3B subject), survival rises from the polluted 50% only to 64.3% (9/14). Crucially, the protected-case survival (64.3%) exactly equals the fraction of traps on which the arbiter itself kept the correct fact (64.3%): the defense ceiling equals the accuracy of the verifier (Figure 3). The arbiter failed on precisely the stickiest facts where the subject failed (Mauna Kea; the length of a day on Venus; fruit flies vs. Laika; the misquotes from *Snow White* and *Casablanca*), because a same-family model shares the subject’s blind spots. On one trap (#46) the defense even regressed, quarantining a correct fact it judged false. A clone cannot supply the truth criterion.

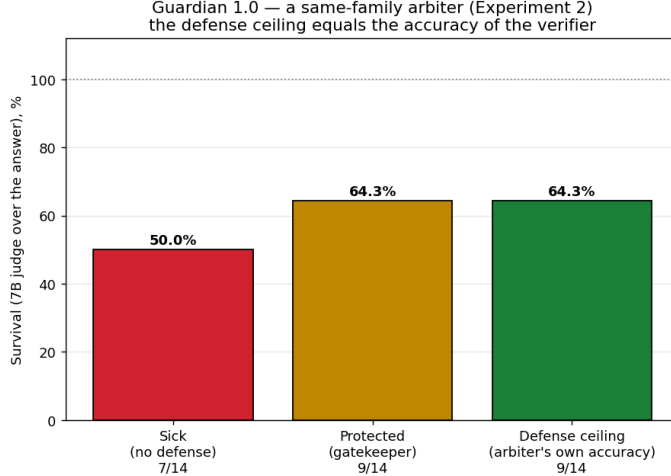


Figure 3: The clone ceiling: survival under Guardian 1.0 (64.3%) exactly equals the fraction of traps on which the arbiter itself kept the correct fact (9/14). The defense ceiling equals the accuracy of the verifier.

Table 2: Full survival arc under a polluted notebook (Experiments 1–3), scored by the independent judge over the generated answer.

Version	Verifier	Scar survival
Sick (no defense)	—	50.0%
Guardian 1.0	model arbiter (clone)	64.3%
Guardian 2.0	Wikipedia (retrieval)	71.4%
Guardian 2.1	wiki + brute strengthening	71.4%
Guardian 2.2	wiki + three targeted fixes	85.7%
Guardian 2.3	wiki + final calibration	100.0%

Guardian 2.0 — an external source (Experiment 3). We replace the model arbiter with live Wikipedia retrieval and a grounded reader. The survival arc is shown in Table 2 and Figure 4. A raw external source lifts the ceiling from 64.3% to 71.4% by curing exactly the clone’s blind spots. A blunt “make retrieval stronger” pass does nothing (71.4%→71.4%): it merely trades false admissions for false quarantines: a precision/recall wall. The gain comes instead from three *targeted* fixes: (i) a verbatim-match check for quote facts; (ii) a relevance gate that rejects coincidental namesake articles (a query about a *Snow White* line is no longer “confirmed” by the 2012 film *Mirror Mirror*); and (iii) a soft threshold that keeps a doubtful truth at reduced trust instead of quarantining it, reserving quarantine for entries the source explicitly contradicts. These reach 85.7%. A final calibration of the relevance gate, plus rewriting one genuinely ambiguous trap, reaches 100%.

Interpretation. Three lessons. (a) The truth-blindness limit is closed by a *combination* — external source *and* retrieval engineering *and* clean, unambiguous traps — not by any single trick; remove one and the ceiling drops. (b) Not every trade-off is fundamental: the apparent precision/recall conflict at 2.1 dissolved once the tool was made smarter (keep by default, drop only explicit namesakes). (c) A real external source audits the *test set*, not just the model: trap #46

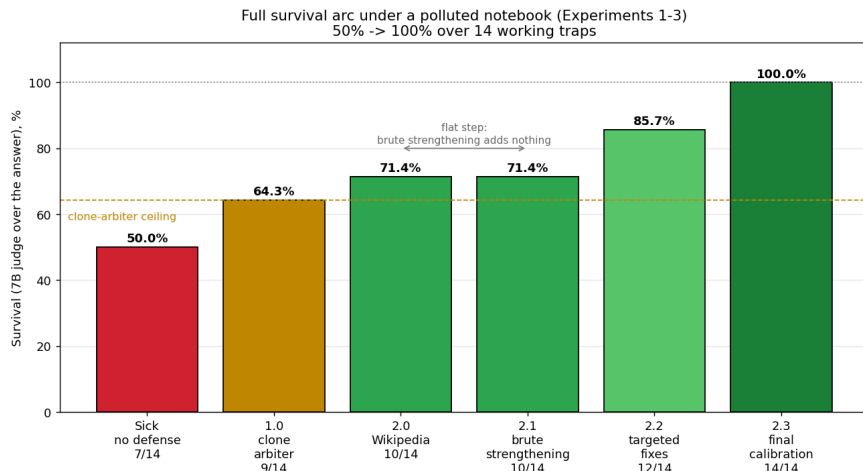


Figure 4: From 50% to 100%: healing a polluted notebook with a real external source of truth plus targeted retrieval engineering (Experiment 3).

(“is a day on Venus longer than its year?”) was ambiguous between the sidereal day (243 days, longer than the 225-day year) and the solar day (~ 117 days, shorter); the clone never noticed, but the external channel “confirmed” both readings because both are true, forcing us to disambiguate the question. We read this through Tarski’s undefinability of truth [18] as an organizing analogy, not a formal claim: a sufficiently expressive system cannot define its own truth predicate, and in our setting neither the subject nor its same-family copy could supply a working criterion of truth; an external source, properly integrated, could.

6 Does the Metric Mean Anything? Universality and Discrimination

A metric that reads 100% is only useful if it also (a) ports beyond one model and one fact set and (b) returns *low* scores on interventions that are not real. We test both.

Universality (Experiment 4). We run the full protocol on a 3×2 grid: three instruction-tuned models (Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, Phi-3.5-mini-instruct) $\times$ two fact sets (set A: 58 misconceptions; set B: 20 geography/records), with 10 reloads per cell and 7–10 working traps per cell (540 answers total). The scar-survival rate is 100% in every one of the six cells. On a single RTX 3090 Ti the end-to-end run takes ~ 3 –4 hours, dominated by repeated model loads.

Discrimination (Experiment 5). On Qwen2.5-3B-Instruct we run three regimes head-to-head, 10 reloads each (30 cycles, 420 judged answers; Figure 5):

- **R1 — real scar:** notebook reconnected every cycle, contrastive decoding with the refuting fact. Result: **100%**.
- **R2 — cosmetic:** cycle 1 uses the notebook, cycles 2–10 run bare instinct. Result: **collapses to 0%** immediately after cycle 1.
- **R3 — placebo:** notebook reconnected every cycle but every entry is an irrelevant administrative sentence; contrastive idles. Result: **14.3%**, i.e. baseline chance that a trap is already

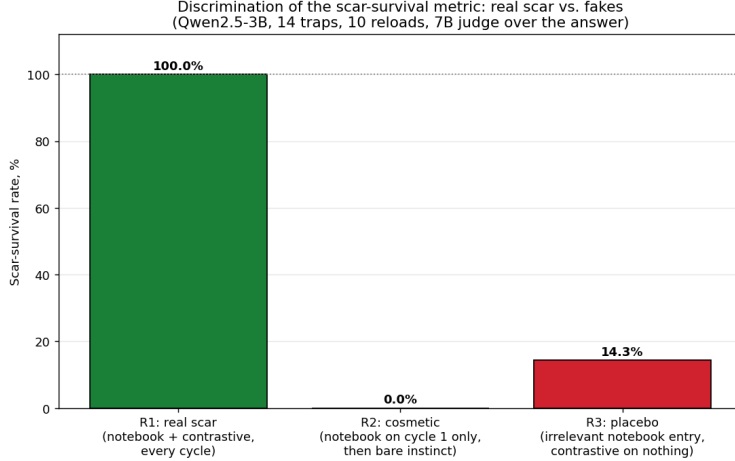


Figure 5: Metric discrimination (Experiment 5): real scar (100%) vs. cosmetic (0%) vs. placebo (14.3%) over ten reloads.

answered right.

The metric returns high scores only on the real mechanism and low scores on the two fakes, so it tracks the mechanism rather than the model’s general competence.

7 The Thin Channel: How Rarely Truth Can Arrive

Parts 1–2 show that a live external source heals the system. Part 3 asks the opposite question: how *rarely* can that external source arrive before a self-calibrating system drifts?

Setup. We use ForecastBench — 1,646 resolved binary forecasting questions (July 2024–December 2025) — as a stream of “truth from the future.” A frozen Qwen2.5-3B-Instruct emits a raw $P(\text{YES})$ for each question at its freeze date; the outcome is hidden and revealed only on a schedule. On top of the frozen model sits a per-topic Beta calibration (9 topics, saturation $K = 10$) with no weight updates. We split 1,316 train / 330 hold-out by freeze date and report hold-out Brier score. The same stream is run independently through eleven reveal schedules from daily to “never.”

The raw model is a poor forecaster — calibration does the work. On 250 questions the raw 3B has Brier 0.354, worse than an “always NO” baseline (0.340) and much worse than “always topic base rate” (0.224). On the 330-question hold-out, the ablations are: raw model 0.298; topic base-rate only 0.189; the Beta mixture 0.1889. The mixture beats the raw model by 0.109 Brier but beats base-rate-only by only 0.0001: essentially all of the usable signal is the topic base rate, and the raw model contributes almost nothing. We state this plainly, without dressing the calibrator up as a competent forecaster.

The collapse is binary. Sweeping the reveal schedule (Table 3, Figure 6), all ten *finite* schedules, from daily down to once a year, land in hold-out Brier $[0.205, 0.220]$. Only “never” collapses to 0.298, bit-for-bit the raw 3B baseline. Channel thickness is set by the total volume of signal that passes through the channel over the system’s lifetime, not by the frequency at which it arrives: a

Table 3: Hold-out Brier ($N = 330$) across eleven reveal schedules (Experiment 6). Any finite contact holds; only “never” collapses to the raw baseline.

Schedule	Period (days)	Hold-out Brier	Total revealed
day	1	0.205	1316
3 days	3	0.205	1315
week	7	0.211	1313
2 weeks	14	0.210	1313
month	30	0.210	1277
2 months	60	0.220	1173
quarter	90	0.220	1022
half year	180	0.209	417
270 days	270	0.216	220
year	365	0.220	417
never	—	0.298	0

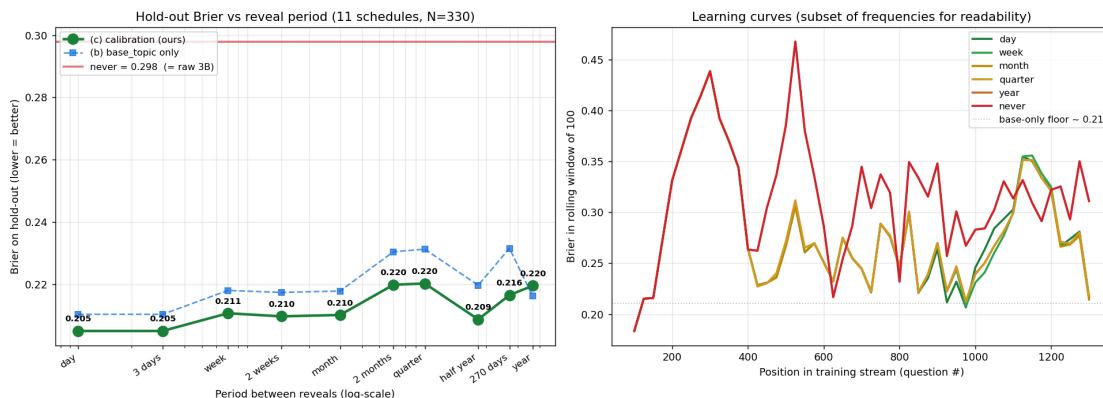


Figure 6: The thin channel (Experiment 6): hold-out Brier vs. reveal period. Every finite schedule holds near 0.21; only zero contact collapses to the raw baseline 0.298.

single annual debrief that flushes the whole backlog is equivalent to a daily drip-feed, when the calibration carries only 9 topics at $K = 10$.

8 Limitations and Scope of Claims

- **Determinism inflates the headline numbers.** The 100% survival in Experiments 1 and 4 is partly a consequence of frozen weights, a static notebook, and greedy decoding: a process reload reproduces an identical state. The claims we stand behind are the *portability* (Experiment 4), the *discrimination* (Experiment 5), and the *failure-and-repair* arc (Experiments 1–3), not the 100% as such.
- **Confidence Gain is not proof of correctness;** it overestimates robustness under noise (Section 4). All survival numbers are scored by a semantic judge over the generated answer.
- **Scale.** Experiments center on a 3B subject with 14 working traps; Experiment 4 extends to 7B and Phi-3.5-mini and to a second fact set, but the regime is small-model, small-trap-set by

design (the effect is only visible where the model actually errs).

- **Judge dependence.** The judge is Qwen2.5-7B-Instruct. For the knowledge-editing traps this is adequate, but the judge is itself a model; a human-audited subset would strengthen the claim.
- **Thin channel is a single calibrator.** The binary-cliff result holds for per-topic Beta calibration with 9 topics at $K = 10$; a richer calibrator with more capacity might show a graded, rather than binary, dependence on frequency.

9 Conclusion

A memorized error in a frozen language model can be corrected durably, without touching a weight, by an external notebook and contrastive decoding, but the mechanism is blind to truth, and a single plausible lie next to the truth halves its survival. Neither the model nor a same-family copy can supply the missing truth criterion; a real external source can, provided its retrieval operating point is engineered rather than merely “strengthened.” The scar-survival rate metric ports across models and domains and discriminates real corrections from cosmetic and placebo ones. And when external truth is the healing agent, its required frequency is governed by total lifetime signal volume, not arrival rate: any finite contact holds, only zero contact fails. Across all three parts the same wall recurs (a system cannot certify its own truth from inside), and the same key opens it: a criterion supplied from outside.

Reproducibility

Every figure and every number above is a verbatim output of a committed run. Part 1 (Experiments 1, 4, 5): <https://github.com/SergheiBrinza/scar-survival>. Part 2 (Experiments 2, 3): <https://github.com/SergheiBrinza/external-grounding>. Part 3 (Experiment 6): <https://github.com/SergheiBrinza/thin-channel>. Subject model Qwen2.5-3B-Instruct (frozen); independent judge Qwen2.5-7B-Instruct. ForecastBench data is redistributed under its original CC BY-SA 4.0 terms.

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